Rekia Fadla

Final-Year Computer Systems Engineering Student at ESI-SBA | Seeking Internship
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SUMMARY

Final-year Computer Systems Engineering student at ESI-SBA. Experienced in building end-to-end IoT systems—from hardware wiring and microcontroller firmware to deployed backends, live dashboards, and applied machine learning pipelines. Additional strong foundations in full-stack development and network security. Seeking a research internship in embedded systems, edge AI, or applied ML.

EDUCATION

Higher School of Computer Science (ESI-SBA), Sidi Bel Abbès, Algeria *2022 – 2027*
Engineering Degree in Computer Science — Systems Engineering (ISI) — Expected 2027.
Baccalaureate in Mathematics — 2022 — Graduated with honors, 17.25/20.

TECHNICAL SKILLS

Programming: C, C++, Python, JavaScript (Node.js), SQL, PL/SQL, R, MATLAB
Embedded & IoT: ESP32, Arduino, Raspberry Pi, ARM, SPI, I^2C, UART, MQTT, Node-RED, embedded Linux (buildroot).
Networking: TCP/IP, VLANs, dynamic routing (RIP, OSPF, BGP), QoS, IPv6, SNMP, Wireshark, Packet Tracer.
AI & Data: scikit-learn, OpenCV, YOLOv12 (ONNX), EasyOCR, NumPy, Weka, PCA, TensorFlow/Keras, TFLite.
Backend & Tools: Node.js, Express, Flask, REST, WebSocket, MongoDB, SQL, Git, Linux, Render, Vercel.

SELECTED PROJECTS

TinyML Environmental Anomaly Detection *Solo*
github.com/roukaiafadla/tinyml-env-anomaly
- Designed an unsupervised autoencoder (101 parameters) to learn joint sensor behavior from DHT11 and PIR data, detecting anomalies via reconstruction error.
- Implemented a full data pipeline: ESP32 flash-logger (C++), Python (TensorFlow) training, and INT8 quantization to a highly compact 3.4 KB model.

Smart Parking — RFID and license-plate access control *Team of 4*
github.com/roukaiafadla/smart-parking • supervised by Prof. A. Rahmoun
- Co-built a distributed access-control system integrating Raspberry Pi 4, Arduino, Flask, and MongoDB with fail-safe barrier behavior.
- Led hardware integration (RFID, sensors) and integrated a two-stage computer vision pipeline (YOLOv12 ONNX + EasyOCR) for real-time plate recognition.

PlantCare — End-to-end IoT irrigation and telemetry system *Solo*
github.com/roukaiafadla/plantcare • live dashboard
- Built a complete IoT system: ESP32 firmware samples soil and climate data, publishing JSON over MQTT.
- Engineered fail-safe local auto-watering logic and built the Node.js/MongoDB backend with a live WebSocket dashboard.

ml-lab — Machine learning algorithms from scratch with NumPy *Solo*
github.com/roukaiafadla/ml-lab
- Implemented linear and logistic regression from scratch (vectorized gradient descent, analytic gradients verified against finite differences).
- Achieved test RMSE of 0.7457 vs 0.7456 for scikit-learn; built reproducible experiments with pytest and CI (GitHub Actions).

ADDITIONAL PROJECTS (FULL-STACK & APPLIED AI)

Wassit (Full-Stack + AI): Hyperlocal marketplace built with React, Node.js, and MongoDB. Integrated Google Gemini API for AI-assisted job posting. (Live • Code)

Shaghaf (MERN + AI): Community platform featuring JWT auth, Cloudinary uploads, and Google Gemini API for AI image understanding. (Live • Code)

Wanderly (Full-Stack): Travel platform with a custom zero-framework Node.js backend, libSQL/Turso database, and CI-tested API. (Live • Code)

PROFESSIONAL EXPERIENCE

Network Engineering Intern — Algérie Télécom, Laghouat *September 2025*
- Analyzed core network infrastructures, fiber-optic transmission centers, and GPON/FTTH access architectures at the national operator's regional headquarters.
- Explored enterprise data systems (Oracle Finance, HR Access) and routing hardware; practiced VLAN and RIP configuration via Cisco Packet Tracer.

LANGUAGES

Arabic (Native) | **French** (B2) | **English** (B2).